CS4340 – Tutorial 2S

Q1 – Use Case Diagram

Diagram (or diagrams) accurately depicting the Technician's interactions with the system.

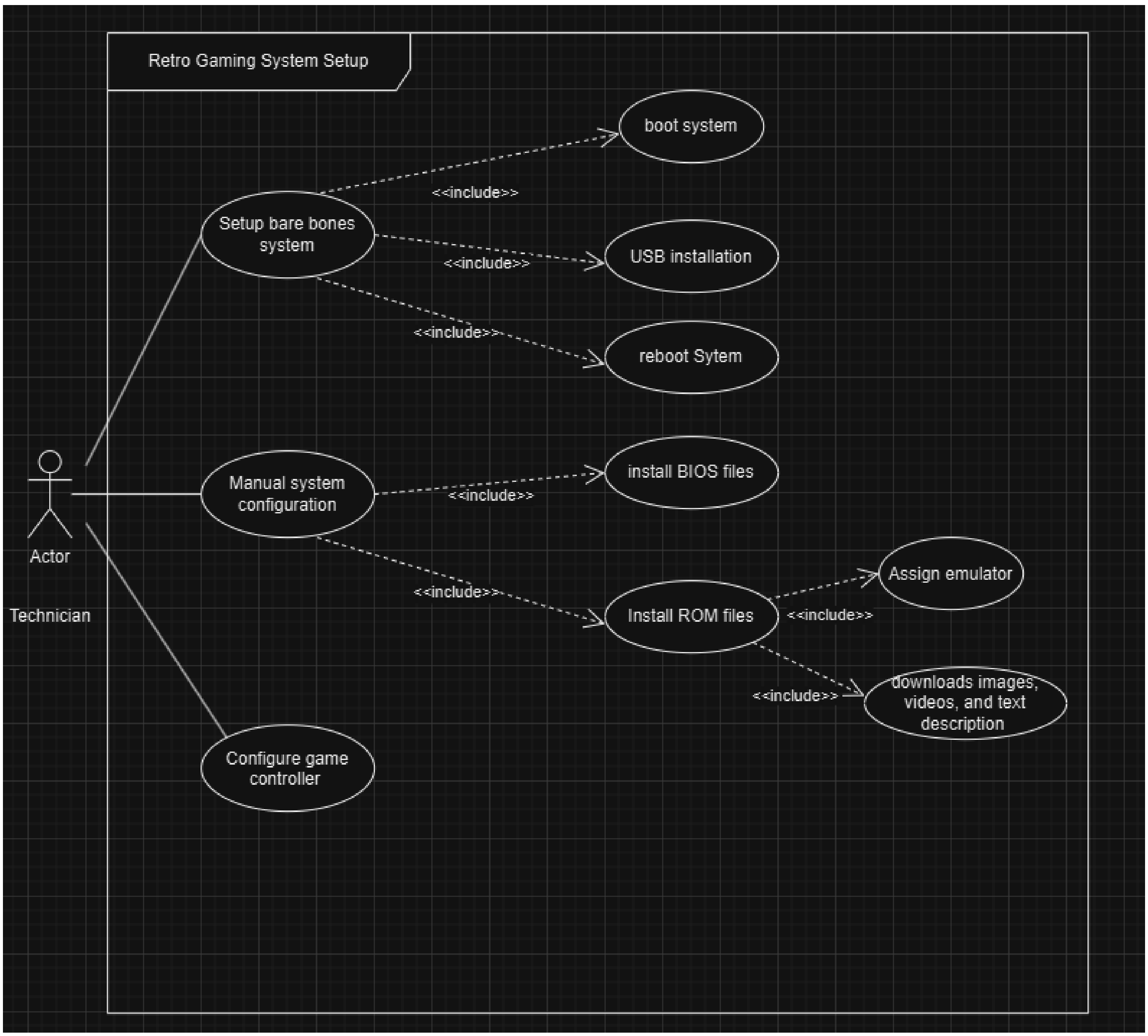
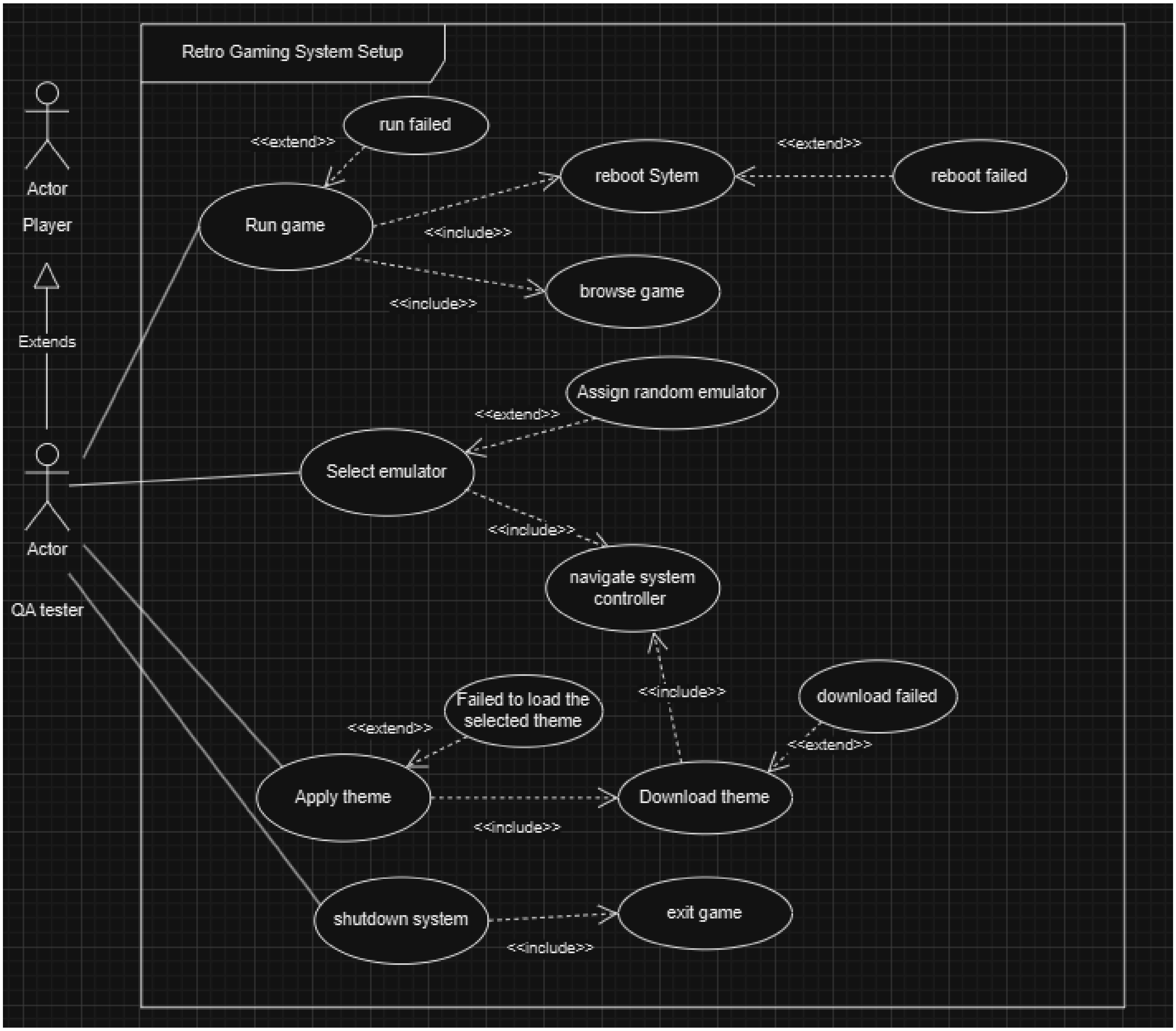


Diagram (or diagrams) accurately depicting the QA Tester's interactions with the system.



Q2 – Coding Best Practice

Reformat/structure the code so that it clearly follows the taught convention i.e. Clearly indicated Class, Methods and Main program sections of code, Code appropriately indented.

```java
import java.util.*;

class troutersion1 {

    public static String WibbleToTrout(int wibble) {
        String trout = "";
        while (wibble > 0) {
            int HalfaWibble = wibble % 2;
            trout = HalfaWibble + trout;
            wibble = wibble / 2;
        }
        return trout;
    }

    public static String WibbleToWombat(int wibble) {
        int HalfaWibble;
        String Wombat = "";
        char WombatBits[] = {'0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'A', 'B', 'C', 'D', 'E',
'F'};
        while (wibble > 0) {
            HalfaWibble = wibble % 16;
            Wombat = WombatBits[HalfaWibble] + Wombat;
            wibble = wibble / 16;
        }
        return Wombat;
    }

    public static String WibbleToMooMoo(int wibble) {
        int HalfaWibble;
        String MooMoo = "";
        char MooMooBits[] = {'0', '1', '2', '3', '4', '5', '6', '7'};
        while (wibble > 0) {
            HalfaWibble = wibble % 8;
            MooMoo = MooMooBits[HalfaWibble] + MooMoo;
            wibble = wibble / 8;
        }
        return MooMoo;
    }

    public static void main(String args[]) {
        System.out.println("The Trout for 10 (Wibble) is: " + WibbleToTrout(10));
        System.out.println("The Trout for 21 (Wibble) is: " + WibbleToTrout(21));
        System.out.println("The Trout for 31 (Wibble) is: " + WibbleToTrout(31));
        System.out.println();

        System.out.println("The Wombat for 10 (Wibble) is: " + WibbleToWombat(10));
        System.out.println("The Wombat for 15 (Wibble) is: " + WibbleToWombat(15));
        System.out.println("The Wombat for 289 (Wibble) is: " + WibbleToWombat(289));
        System.out.println();

        System.out.println("The MooMoo for 8 (Wibble) is: " + WibbleToMooMoo(8));
        System.out.println("The MooMoo for 19 (Wibble) is: " + WibbleToMooMoo(19));
        System.out.println("The MooMoo for 81 (Wibble) is: " + WibbleToMooMoo(81));
        System.out.println();
    }
}
```

Informative/sensible variable naming, following a consistent formatting convention.

```java
import java.util.*;

class ConversionUtility {

    public static String convertToBinary(int decimalNumber) {
        String binaryRepresentation = "";
        while (decimalNumber > 0) {
            int remainder = decimalNumber % 2;
            binaryRepresentation = remainder + binaryRepresentation;
            decimalNumber = decimalNumber / 2;
        }
        return binaryRepresentation;
    }

    public static String convertToHexadecimal(int decimalNumber) {
        String hexadecimalRepresentation = "";
        char[] hexadecimalCharacters = {'0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'A', 'B', 'C',
'D', 'E', 'F'};
        while (decimalNumber > 0) {
            int remainder = decimalNumber % 16;
            hexadecimalRepresentation = hexadecimalCharacters[remainder] + hexadecimalRepresentation;
            decimalNumber = decimalNumber / 16;
        }
        return hexadecimalRepresentation;
    }

    public static String convertToOctal(int decimalNumber) {
        String octalRepresentation = "";
        char[] octalCharacters = {'0', '1', '2', '3', '4', '5', '6', '7'};
        while (decimalNumber > 0) {
            int remainder = decimalNumber % 8;
            octalRepresentation = octalCharacters[remainder] + octalRepresentation;
            decimalNumber = decimalNumber / 8;
        }
        return octalRepresentation;
    }

    public static void main(String[] args) {
        System.out.println("The binary representation of 10 is: " + convertToBinary(10));
        System.out.println("The binary representation of 21 is: " + convertToBinary(21));
        System.out.println("The binary representation of 31 is: " + convertToBinary(31));
        System.out.println();

        System.out.println("The hexadecimal representation of 10 is: " + convertToHexadecimal(10));
        System.out.println("The hexadecimal representation of 15 is: " + convertToHexadecimal(15));
        System.out.println("The hexadecimal representation of 289 is: " + convertToHexadecimal(289));
        System.out.println();

        System.out.println("The octal representation of 8 is: " + convertToOctal(8));
        System.out.println("The octal representation of 19 is: " + convertToOctal(19));
        System.out.println("The octal representation of 81 is: " + convertToOctal(81));
        System.out.println();
    }
}
```

The final reformatted/renamed code will need to be documented, to indicate your understanding of the entire program.

```java
/*************************************************
 * Class: ConversionUtility                      *
 *                                               *
 * This program performs conversions of decimal  *
 * numbers into binary, hexadecimal, and octal   *
 * representations. The user is provided with    *
 * methods to convert a decimal number into      *
 * each of these formats and view the results.   *
 *                                               *
 * Methods:                                      *
 * --------                                      *
 * 1. convertToBinary: Converts a decimal number *
 *    into its binary representation.            *
 *                                               *
 * 2. convertToHexadecimal: Converts a decimal   *
 *    number into its hexadecimal representation. *
 *                                               *
 * 3. convertToOctal: Converts a decimal number  *
 *    into its octal representation.             *
 *                                               *
 * The main method demonstrates the functionality *
 * of the program by using predefined test cases. *
 *************************************************/

import java.util.*;

class ConversionUtility {

    /**
     * Converts a decimal number to its binary representation.
     *
     * @param decimalNumber The decimal number to be converted.
     * @return A string representing the binary equivalent of the input.
     */
    public static String convertToBinary(int decimalNumber) {
        String binaryRepresentation = ""; // Stores the binary result
        while (decimalNumber > 0) {
            int remainder = decimalNumber % 2; // Calculate the remainder when divided by 2
            binaryRepresentation = remainder + binaryRepresentation; // Append to the result
            decimalNumber = decimalNumber / 2; // Update the number for the next iteration
        }
        return binaryRepresentation;
    }

    /**
     * Converts a decimal number to its hexadecimal representation.
     *
     * @param decimalNumber The decimal number to be converted.
     * @return A string representing the hexadecimal equivalent of the input.
     */
    public static String convertToHexadecimal(int decimalNumber) {
        String hexadecimalRepresentation = ""; // Stores the hexadecimal result
        char[] hexadecimalCharacters = {
            '0', '1', '2', '3', '4', '5', '6', '7',
            '8', '9', 'A', 'B', 'C', 'D', 'E', 'F'
        }; // Array of hexadecimal characters
        while (decimalNumber > 0) {
            int remainder = decimalNumber % 16; // Calculate the remainder when divided by 16
            hexadecimalRepresentation = hexadecimalCharacters[remainder] + hexadecimalRepresentation;
            decimalNumber = decimalNumber / 16; // Update the number for the next iteration
        }
        return hexadecimalRepresentation;
    }

    /**
     * Converts a decimal number to its octal representation.
     *
     * @param decimalNumber The decimal number to be converted.
     * @return A string representing the octal equivalent of the input.
     */
    public static String convertToOctal(int decimalNumber) {
        String octalRepresentation = ""; // Stores the octal result
        char[] octalCharacters = {'0', '1', '2', '3', '4', '5', '6', '7'}; // Array of octal
characters
        while (decimalNumber > 0) {
            int remainder = decimalNumber % 8; // Calculate the remainder when divided by 8
            octalRepresentation = octalCharacters[remainder] + octalRepresentation;
            decimalNumber = decimalNumber / 8; // Update the number for the next iteration
        }
        return octalRepresentation;
    }

    /**
     * Main method to demonstrate the functionality of the program.
     *
     * It uses predefined test cases to convert decimal numbers
     * into binary, hexadecimal, and octal formats and prints the results.
     */
    public static void main(String[] args) {
        // Demonstrate binary conversion
        System.out.println("The binary representation of 10 is: " + convertToBinary(10));
        System.out.println("The binary representation of 21 is: " + convertToBinary(21));
        System.out.println("The binary representation of 31 is: " + convertToBinary(31));
        System.out.println();

        // Demonstrate hexadecimal conversion
        System.out.println("The hexadecimal representation of 10 is: " + convertToHexadecimal(10));
        System.out.println("The hexadecimal representation of 15 is: " + convertToHexadecimal(15));
        System.out.println("The hexadecimal representation of 289 is: " + convertToHexadecimal(289));
        System.out.println();

        // Demonstrate octal conversion
        System.out.println("The octal representation of 8 is: " + convertToOctal(8));
        System.out.println("The octal representation of 19 is: " + convertToOctal(19));
        System.out.println("The octal representation of 81 is: " + convertToOctal(81));
        System.out.println();
    }
}
```